CSE 213 Microcontroller Programming Quiz Soruları

Bu doküman, Quiz 2 ve Quiz 3'e ait soruları, şıkları ve doğru cevapları içermektedir.

Quiz 2 (12 Aralık 2024)

25G CSE 213 Microcontroller Programming Quiz 2

Q1

Write an 8065 assembly code to find the second largest element in an array of integers stored in memory. For arithmetic operations use BL for the largest element and DL for the second largest element. The helper labels are provided. (Integer value of 0 marks the end of the array.)

Yorum: * Find largest = 15pts * Find second largest = 0pts * Code Effort = 5pts

Cevap: (Açık uçlu soru olduğu için cevap kodu burada verilmemiştir.)

Q2

What is the value of AL after the execution of given code?

```
 org 100h

      MOV SI, 5
      MOV BX,2
      MOV CX,5
      MOV AL, Dots[BX][SI]

ret
Dots DB   00011000b, 00111100b, 01111000b, 00111100b, 00011000b  ;
        DB    01111110b, 00001001b, 00001001b, 00001001b, 01111110b  ;
        DB    00111110b, 01000001b, 01000001b, 01000001b, 00100010b  ;
        DB    01111111b, 01001001b, 01001001b, 01001001b, 00110110b  ;
        DB    01111111b, 01001001b, 01001001b, 01001001b, 00110110b  ;
        DB    01111110b, 00001001b, 00001001b, 00001001b, 01111110b  ;
        DB    01111111b, 00001001b, 00001001b, 00001001b, 01110110b  ;
        DB    00011000b, 00111100b, 01111000b, 00111100b, 00011000b  ;
```

Şıklar: a. `01110110b` b. `01001001b` c. `00011000b` d. `00001001b` e. `01001001b`

Doğru Cevap: `00001001b`

Q3

In the given assembly code, what is the final value of BX register in decimal?

```
 org 100h

mov bx, 0
mov cx, 5
k2:
add bx, 1
mov si,cx
mov cx, 6
k3:
add bx, 1
loop k3
mov  cx,si
loop  k2
ret
```

Şıklar: a. 60 b. 30 c. 11 d. 35 e. 40

Doğru Cevap: 35

Q4

We want to check if a number is even and store the result in "iseven" variable. Is there any problem with the given code? If so pick your appropriate solution.

```
 org 100h
mov bl,2
mov ax, number
div bl
cmp ah,0
je even
ret
even:
mov bl,1h
mov iseven,bl
ret
number dw 24h
iseven db 0
```

Şıklar: a. `bl` should be `bx` b. It works as it is. c. There is no need for two `ret` instructions d. `cmp ah,0` should be changed with `cmp dx,0` e. `Number` should be `db`

Doğru Cevap: It works as it is.

Q5

Considering the code below what would be the final state of `vec3` in decimal?

```
 org 100h
jmp start
vec1 db 1, 2, 5, 6
vec2 db 3, 5, 6, 1
vec3 db 0, 0, 0, 0
start:
lea si, vec1
lea bx, vec2
lea di, vec3
mov cx, 4
mov ah,0

sum:
    mov al, [si+1]
    add al, [bx+2]
    add [di+3], al
    inc ah
    loop sum
ret
```

Şıklar: a. 7, 10, 14, 11 b. 0, 0, 0, 32 c. 4, 7, 11, 7 d. 0, 0, 8, 0 e. 0, 0, 32, 0

Doğru Cevap: 0, 0, 0, 32

Quiz 3 (19 Aralık 2024)

25G CSE 213 Microcontroller Programming Quiz 2 (Tarih farklı, başlık aynı)

Q1

Assume that `RESULT` and `NUMBERS` variables are correct. We are having trouble displaying the correct number. What operation would you add to `------`?

```
DISPLAY_RESULT:
    MOV AH,0
    MOV AL,RESULT
    DIV BL
    ---------
    MOV SI, AX
    MOV AL, NUMBERS[SI]
    MOV DX, 2036H
    OUT DX, AL
```

Şıklar: a. `MOV DX,2030H` b. `INC SI` c. `MOV BL,AL` d. `XOR AH,AH` e. `MOV AL,0`

Doğru Cevap: `XOR AH,AH`

Q2

What is the value of AX after running the following program, given that the array `numbers` contains `{10, 25, 18, 42, 30, 5}`?

```
 org 100h
mov cx, 6
lea si, numbers
mov ax, [si]
add si, 2
dec cx
find:
    mov bx, [si]
    cmp ax, bx
    jl update_ax
    jmp skip_update
update_ax:
    mov ax, bx
skip_update:
    add si, 2
    loop find
ret
numbers dw 10, 25, 18, 42, 30, 5
```

Şıklar: a. 5 b. 25 c. 42 d. 10 e. 18

Doğru Cevap: 42

Q3

What is the final value of AX?

```
org 100h

LEA SI, array
MOV AX, 0
MOV CX,5
SUM_LOOP:
    MOV BX, [SI]
    CMP BX, 0
    JE END_LOOP
    MOV DX, 0
    DIV BX, 5
    CMP DX, 0
    JNZ SKIP
    ADD AX, BX
SKIP:
    INC SI
    JMP SUM_LOOP
END_LOOP:
ret
array dw 12, 15, 25, 32, 44, 10, 0
```

Şıklar: a. 128 b. 40 c. 15 d. 50 e. register overflow

Doğru Cevap: register overflow

Q4

What does the memory zero terminated array contain after running the following program, if the input array is `{1, 2, 3, 4, 5, 6, 0}`?

```
org 100h
lea si, array
lea di, array

find_length:
    mov al,[di]
    cmp al, 0
    je reverse_loop
    inc di
    jmp find_length

reverse_loop:

    cmp si, di
    jge end_reverse
    mov al, [si]
    xchg al, [di]
    mov [si], al
    add si, 1
    sub di, 1
    jmp reverse_loop
end_reverse:
ret
array db 1, 2, 3, 4, 5, 6, 0
```

Şıklar: a. 0, 1, 2, 3, 4, 5 b. 1, 2, 3, 4, 5, 6 c. 0, 6, 5, 4, 3, 2 d. 2, 3, 4, 5, 6, 0 e. 6, 5, 4, 3, 2, 1

Doğru Cevap: 0, 6, 5, 4, 3, 2

Q5

Is there anything wrong with the code for achieving the task? If so how would you fix it?

Şıklar: a. we need to change (`xchg al, [di]` to `xchg [si], [di]`) b. We need to change (`add si,1` to `add si,2`) and (`dec di ,1` to `add di,2`) c. It works as intended d. We must add a new label with `dec di` before the `reverse_loop` and change `je reverse_loop` accordingly. e. change `jge end_reverse` with `jg end_reverse`

Doğru Cevap: We must add a new label with `dec di` before the `reverse_loop` and change `je reverse_loop` accordingly.

Q6

What is the value of AX after running the following code snippet, assuming the array `nums` contains `{1, 2, 3, 4, 5, 6, 7}`?

```
 org 100h
mov cx, 7
lea si, nums
xor ax, ax
sum_odd:
    mov bx, [si]
    test bx, 1
    jz skip_add
    add ax, bx
skip_add:
    add si, 2
    loop sum_odd
ret
nums dw 1, 2, 3, 4, 5, 6, 7
```

Şıklar: a. 28 b. 12 c. 18 d. 16 e. 10

Doğru Cevap: 16